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S410 WELDING WORKS

1. SCOPE OF WORK

- 1.1 All welding for steel constructions shall conform to the requirements of this section and shall be to the Bidder's work included in Part 4, Subsection "Summary of Work" and Subsections S-001 "General Design Requirement of Structure, Civil and Building Works".
- 1.2 Welding covered in this Specification shall include shop and field welding of the following, as indicated in various technical specifications included with the bid document or as required for the Bidder's work:
 - 1.2.1 Structural steel, galleries, embedded steel, auxiliary steel and miscellaneous steel, including pile splices.
 - 1.2.2 Welding studs.
 - 1.2.3 All other work as indicated.

2. CODES AND STANDARDS

- 2.1 The Bidder shall comply with the requirements of Part 4, Subsection 2.5 "Codes and Standards".
- 2.2 The codes and standards of the following organizations extracted from the Part 4, Subsection 2.5 "Codes and Standards" are specifically applicable to the design, manufacture and testing of the work included in this Specification:
 - JIS Japanese Industrial Standards
 - ASTM American Society for Testing and Materials
 - AWS American Welding Society.

3. MATERIALS

3.1 General Provisions

All materials shall conform to the requirements of this Clause 3 unless otherwise indicated in the Bid Document.

3.2 Stud Welding Materials

Materials for stud welding shall conform to the requirements of AWS D1.1.

3.3 Electrodes

Except as otherwise specified, all welding electrodes shall conform to the applicable provisions of the AWS Structural Welding Code AWS D1.1 for steel with special properties, such as resistance to brittle fracture, corrosion-resistant steels, stainless clad steels and stainless steels, the recommendations of both the steel and electrode manufacturers shall be secured for each specified steel and each specific use.

4. DETAILED REQUIREMENTS

- 4.1 Complete penetration welds shall be used for all groove welding unless otherwise specifically indicated. All double welded joints shall be back gouged to sound metal before welding the second side.
- 4.2 Fillet welds across flanges of any steel member will not be permitted unless specifically indicated on Bidder's design drawings; any undercutting or damage that will reduce the flange area will not be permitted.
- 4.3 Preheat and interpass temperatures shall conform to the requirements of AWS D1.1 and shall be as required for the thickest part of the base metal being welded.
- 4.4 The maximum preheat and interpass temperature for austenitic stainless steel shall be 175 °C.
- 4.5 Except for stainless steel joints or stainless to carbon steel joints, the austenitic stainless steel filler metal used shall contain a minimum of 5 % ferrite.
- 4.6 The use of wide welds to overcome poor fit-up will not be permitted.
- 4.7 Surfaces and edges to be welded shall be properly prepared according to AWS D1.1, prior to welding. Unless specifically reviewed by the Owner, the only coating which does not have to be removed before welding is deoxaluminite.

- 4.8 In welding stainless steel, the carbon steel side shall be welded first.
- 4.9 The following materials shall not be welded unless otherwise indicated:
- Steel reinforcing bars (re-bars).
 - Nuts and bolts.
 - Any carbon or alloy steels containing more than 0,35% carbon.
- 4.10 The filling in by welding of unused bolt holes is not permitted.

5. PERFORMANCE

- 5.1 All welding including stud welding shall conform to the requirements of AWS D1.1, unless otherwise indicated in the related Indonesian Standards. Welding shall be performed using Shielded Metal Arc or Submerged Arc Welding Process.
- 5.2 The following welding processes are specifically prohibited:
- Fuel Gas
 - Electro slag
 - Electro gas-Gas Metal-Arc (short circuiting transfer)
 - Flux Cored-Arc (without external gas shielding)

6. INSPECTION AND EXAMINATION

6.1 General Provisions

- 6.1.1 All welding shall be inspected and examined by Bidder's AWS certified welding inspector in accordance with this Clause 6 and AWS D1.1, unless otherwise indicated in Indonesian Standard Qualification of Welder. Inspection and examination report shall be submitted to the Owner.
- 6.1.2 Have each welding operator stamp his identifying mark or number on all work done by him (except on handrails) as requested by inspector.
- 6.1.3 When inspector believes a welding operator's work is not up to required standards he may prohibit welding operator from continuing work pending results of re-qualification tests.
- 6.1.4 The Owner may, during the course of the work, review the various phases of the work performer by the Bidder, at the Bidder's plant and at the site, for compliance with all requirements of this specification. The Owner reserves the right to approve facilities used for inspection and examination.
- 6.1.5 Radiographic inspection shall be performed by inspecting laboratory retained by Bidder. Such inspection will conform to the applicable requirements of AWS D1.1.

6.2 Extent And Methods Of Examination

The extent and methods of non destructive examination for welding required for all work shall conform to the requirements of Table S.410-1, and all welds shall be 100 % visually inspected.

Table S.410-1
Extent and Methods of Nondestructive Examination for
Welding for All Steel Structure Work

Item	Extent of examination	Method Examination	Remarks
1. <u>Structural steel</u>			
1.1 Built-up girders and columns:			
a. All complete penetration groove welds.	100 %	Radiograph	Welds impractical to be radio graphed, as determined by the Owner, shall be examined 100% by both Ultrasonic and either Magnetic Particle or Liquid Penetrant method.
b. All other welds joining flanges, webs, plates, stiffeners, connections, base plate details, etc.	100 %	Visual	
1.2 Rolled beams and columns:			
a. All complete penetration groove welds.	100 %	Radiograph	Same as 1.1a
b. All other welds joining flanges, webs, plates, stiffeners, connections, base plate details, etc.	100 %	Visual	

Item	Extent of examination	Method Examination	Remarks
2. <u>Tanks, Ductwork etc</u> 2.1 All complete penetration groove welds	a. 100% for the first 3 m. of welds, in horizontal and vertical welds respectively, for each welder or welding operator. (Any weld that fails to meet the acceptance standard shall be replaced and reexamined 100%)	Radiograph	a. Welds impractical to be radio graphed, as determined by the Owner, shall be examined by both Ultrasonic and either Magnetic Particle or Liquid Penetrant Methods
	b. Thereafter, 1% examination 30 cm long for each additional 33m of welds. (If any spot fails to meet the acceptable standard, 2 additional spots shall be examined at locations at least 30 cm on each side from failure spot. If either of the 2 additional spots examined fails to meet the acceptance standard the entire unit of the 33 m welds shall be completely examined).	Spot Radiograph	b. The 1% spot radio graphed shall include 25% of junctions of vertical and horizontal welds.
2.2 Fillet welds between plates and stiffeners	100 %	Visual	
2.3 Welds for supporting structural steel	Same as Item 1	Same as Item 1	Same as Item 1

Item	Extent of examination	Method Examination	Remarks
3. <u>Welding Stud</u>			
3.1 Composite beam shear connectors	a. The first two studs welded on each beam, through metal deck where applicable, after being allowed to cool, shall be bent to an angle of 30° from their original axes by being struck with a hammer. (If failure occurs in either stud, the procedure shall be corrected and two more shall be welded to the beam and tested. If either of the second pair of studs fails, additional welding shall be continued on separate plates until two consecutive studs are tested and found to be satisfactory.)	Bend Test	Additional acceptance standard indicated in AWS D1.1
	b. 100% of all stud	Visual	Any stud which does not show a 360° weld under visual examination is subject to the bend test
3.2 All other studs	a. Before the start of welding operations on each day for each welding gun and associated power source two studs shall be welded to separate material in the same general position (flat, vertical, overhead sloping) and of similar thickness and material. After being allowed to cool, each stud shall be bent to an angle of 30° from its original axis with a hammer. (If failure occurs in either stud, the procedure shall be corrected and repeated until two consecutive studs have been successfully welded and tested before any studs are welded to the member).	Bend Test	Additional acceptance standard indicated in AWS D1.1

Item	Extent of examination	Method Examination	Remarks
	b. 100 % of all studs	Visual	Any stud which does not show a 360° weld under visual examination is subject to the bend or torque test.
	c. At least one stud in every 100 shall be bent to an angle of 15° from its original axis by being struck with a hammer. If threaded, the stud shall be torque tested with a calibrated torque wrench. (If the stud fails, the procedures shall be checked in accordance with (a) above, and two more exiting studs shall be bend or torque tested. If either of these two studs fails, all the studs represented the tests shall be tested.)	Bend Test (Torque Test for threaded studs)	Additional acceptance standard indicated in AWS D1.1

- Notes:
1. Extent of examination (1%, etc.) shall be applied equally to shop welding and field welding.
 2. Magnetic Particle method is for ferrite material, and Liquid Penetrant method is for austenitic materials.

6.3 Requirements For Examination

Non-destructive examination for welding for all work shall conform to the following requirements unless otherwise indicated :

6.3.1 Radiographic Examinations

6.3.1.1 Radiographic examination shall conform to the requirements of this Clause 6.3.1 and to all applicable requirements of the following:

- a. AWS : AWS D1.1
- b. ASTM : E94 and E142

6.3.1.2 X-Ray or Gamma Ray shall be used in radiography and the quality level shall be 2-2T as specified in ASTM E142, unless otherwise indicated.

a. Radiographic Films

a.1 The double film technique shall be used to obtain two films at each radiographic location. Single film viewing shall be use interpretation.

a.2 Radiographic films shall be of fine grain, high contrast, type 1 or 2 Lead intensifying screens shall be used. Film destiny shall be 2.0 minimums for single viewing or 2.6 minimums for composite viewing of a double film exposure. Each film for composite viewing shall have a minimum density of 1.3 as determined by either film density strips or densitometry. The maximum density shall be 3.8 for either single or composite viewing. The quality of film and the steps of processing shall be such as to retain the best possible definition during storage.

a.3 Any films not acceptable to Owner will be rejected; and Bidder shall repair and radiograph again those portions of the work covered by these films until satisfactory are obtained at no cost to the Owner.

6.3.2 **Ultrasonic Examination**

Shall conform to all applicable requirements of the following standards:

a. AWS : AWS D1.1

b. ASTM: E164

6.3.3 **Magnetic Particle Examination**

Shall conform to all applicable requirements of the following standards:

a. AWS : AWS D1.1

b. ASTM: E709

6.3.4 **Liquid Penetrant Examination**

Shall conform to all applicable requirements of the following standards:

- a. AWS : AWS D1.1
- b. ASTM : E165

6.3.5 **Visual Examination**

Shall conform to all applicable requirements of AWS D1.1.

6.4 **Acceptance**

6.4.1 Welds meeting the acceptance standards indicated in AWS D1.1 will be accepted.

6.4.2 Any welds failing to meet the indicated in AWS D1.1 Standard shall be repaired or replaced and re-examined 100 % until they meet the requirements. The Bidder shall perform additional examination in accordance with Table S.410-1 and AWS D1.1, with no additional cost to the Owner.

6.4.3 Acceptance of work by the Owner shall not relieve the Bidder from full responsibility for the work.

7. **QUALITY ASSURANCE PROGRAM**

7.1 **General Procedure**

7.1.1 The Bidder shall submit, after award of contract, two copies of General Procedures. These procedures shall include, but shall not be limited to, all of the following:

7.1.1.1 Selection of welding methods, equipment and materials.

7.1.1.2 Control of personnel proposed to be used for this work, and quality control of workmanship.

7.1.1.3 Control of Bidder's welding inspection and examination.

7.1.1.4 Types of weld and welded joints.

7.1.1.5 Weld profiles and weld sizes.

7.1.1.6 Size, classifications and material specifications for all electrodes proposed for use.

- 7.1.1.7 Methods of groove and edge preparation.
- 7.1.1.8 Length and spacing of intermittent welds.
- 7.1.1.9 Methods and/or sequence of welding to avoid distortion and to minimize residual shrinkage stresses.
- 7.1.1.10 Number and travel speed of passes for each weld.
- 7.1.1.11 Procedures for preheating base metals, and minimum preheat and interpass temperatures required.
- 7.1.1.12 Stud welding, including welding shear connector studs to beams through metal deck.

7.2 Qualification Of Welders And Welding Operations

- 7.2.1 All welders and welding operators employed on any welding work shall be qualified by an approved testing laboratory in accordance with AWS D1.1 or Indonesian Standard Regulation for Welder. Laboratories for testing welders and welding operators shall be of recognized standing. All costs in connection with this qualification shall be borne by Bidder.
- 7.2.2 These qualifications (including re-qualification, if any) test reports and certificates shall be available before the start of welding and shall be submitted for approval to the Owner.

7.3 Quality Assurance/Quality Control (Qa/Qc)

- 7.3.1 Quality assurance/quality control shall conform to the requirements of Part 4, Subsection 2.7, Clause 4.
- 7.3.2 The quality control procedures shall include but not be limited to the following:
 - Material certificates
 - Welding
 - Non-destructive testing
 - Shop test & inspection procedures

7.3.3 The quality control documentation shall include but not be limited to the following:

- Material certificates
- Welding procedures
- Shop test and inspection reports

8. WELDS IMPRACTICAL TO RADIOGRAPH

8.1 Welds which are considered impractical to be radio graphed are those where the joint detail does not permit radiographic examination or the radiography will not yield meaningful result.

8.2 The absence of suitable radiographic equipment shall not be justifications for the alternate examination.

8.3 Examination by the methods the alternate methods shall require permission of the Owner.

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